UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2010 question paper

for the guidance of teachers

5070 CHEMISTRY

5070/41 Paper 4 (Alternative to Practical), maximum raw mark 60

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1 **(a)** blue (1)

(b) blue (1)

(c) **(i)** (light) blue ppt. (1)

(ii) ppt. dissolves (1) dark blue solution (1) [5]

2 **(a)** **(i)** B, reference to metal, positive ions, reduction occurs, –ve electrode etc. (1)

(ii) copper (1)

(b) **(i)** hydrogen (1)

(ii) 'pops' in a flame (1)

(c) **(i)** oxygen (1)

(ii) relights a glowing splint (1)

(d) **(i)** blue to colourless (1)

(ii) copper is being removed from the solution etc./concentration decreased (1) [8]

3 **(a)** **(i)** 3.85 g (1)

(b) **(i)** anhydrous/dehydrated (1)

(ii) 2.30g (1)

(iii) 1.55g (1)

(c) **(i)** 152

(ii) 18 (1)

(d) **(i)** 0.0151 (1)

(ii) 0.086 (1)

(e) x = 5.70 (1) $FeSO_4.6H_2O$ (1) [9]

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4 **(c)** [1]

5 **(b)** [1]

6 **(c)** [1]

7 steeper (1) horizontal line 2 × height (1) [2]

8 **(a)** pink to colourless (1)

(b)

26.6 or 26.5	39.4	30.2
0.0	13.6	4.2
26.6 or 26.5	25.8	26.0

[Mark rows or columns to the benefit of the candidate. One mark for each correct row or column] (3)
Mean value 25.9 cm^3 (1)

(c) 0.0025 (1)

(d) 0.0025 (1)

(e) 0.0965 (1)

(f) 88 (1)

(g) **(i)** n = 3 (1)

(ii) C_3H_7COOH (1)

(h) **(i)** $C_3H_7COOC_2H_5$ (1)

(ii) esters/alkanoates (1) [13]

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9 **(a)** colourless solution (1)

(b) **(i)** aq. NaOH (1) white ppt. (1)

(ii) excess aq. NaOH, ppt. dissolves (1)

(c) **(i)** white ppt. (1)

(ii) ppt. dissolves (1)

(d) HNO_3/aq. $AgNO_3$ (2) white ppt. (1) [9]

10 **(a)** 27, 31, 32 (1)

(b) all points plotted correctly (1)
Two intersecting straight lines (2)

(c) **(i)** 34° (1)

(ii) 22 cm^3 (1)

(d) **(i)** $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$ (1)

(ii) calculation based on moles and mole ratio (1)
0.45 mol/dm^3 (1)

(e) sulfuric acid in excess (1) cools the solution (1) [11]

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